FitBuddy

Technology Stack

Proposed stack from the project brief | Local development version

Application layers

Frontend

HTML and CSS - page structure and responsive styling.
Jinja2 - dynamic server-rendered templates: input, results, and admin pages.

Backend

Python - application language.
FastAPI - web routes, form handling, validation, and API documentation.
Uvicorn - local ASGI server for running the FastAPI app.

AI services

Google Gemini API - generate and update workout plans (brief names Gemini 1.5 Pro).
Gemini Flash - concise nutrition or recovery suggestions (as described in the brief).
Google Generative AI Python SDK - connect the Python app to Gemini.

Data storage

SQLite - local relational database for user and plan records.
SQLAlchemy - Python ORM for database models and queries.

Supporting tools

python-multipart - receive HTML form submissions in FastAPI.
python-dotenv (optional) - load local environment settings such as API keys.
Git and GitHub - version control and phase-wise project submission.

Typical request flow

Browser form -> FastAPI route -> Gemini API and SQLite -> Jinja2 result page

Configuration and security

Keep the Gemini API key in an environment variable; do not place secrets in code or GitHub.
Use a virtual environment and install dependencies from requirements.txt.
Limit admin data access to authorized users and avoid exposing personal data unnecessarily.

Stack reflects the supplied project brief; verify current SDK/model names and compatibility before implementation.